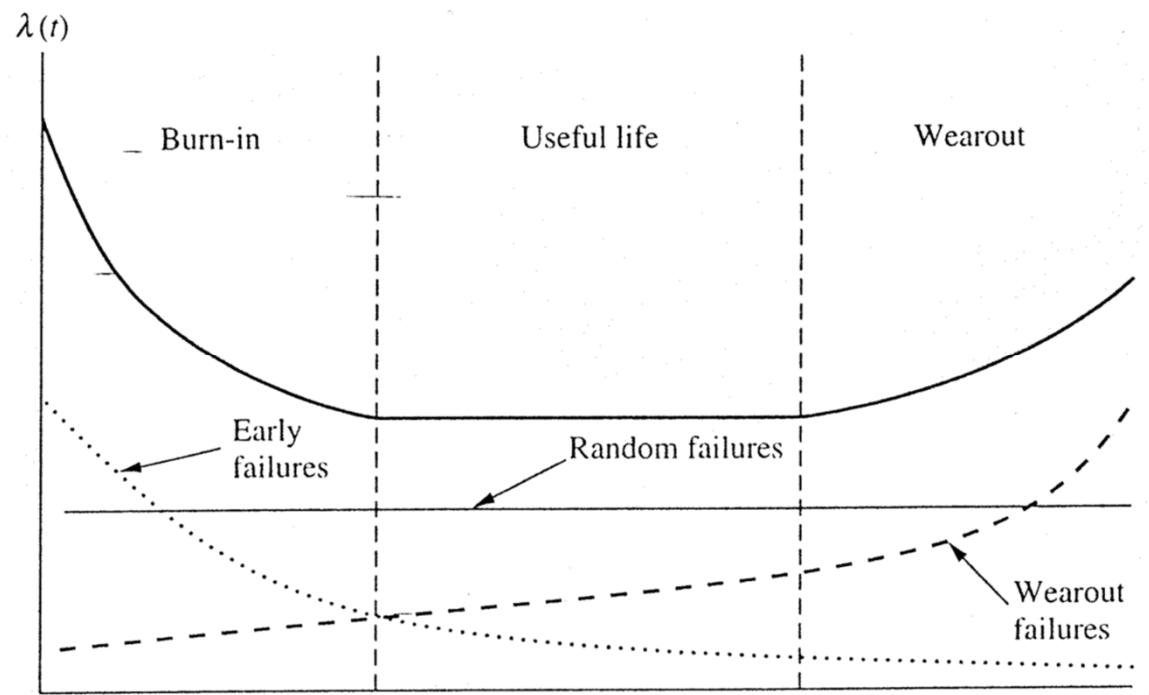


Bathtub Curve: An important form of hazard rate function is commonly referred to as the **bathtub curve** because of its shape. Systems having this hazard rate function experience decreasing failure rate (DFR) early in their life cycle (infant mortality), followed by a nearly constant failure rate (CFR) (useful life), followed by an increasing failure rate (IFR) (wear out).



Stage/ period	Characterized by	Caused by	Reduced by
Burn-in	DFR	Manufacturing defects: welding flaws, cracks, defective parts, poor quality control, contamination, poor workmanship	Burn-in testing, screening, quality control, acceptance testing.
Useful life	CFR	Environment, random loads, human error, act of God, chance event	Redundancy, excess strength.
Wear-out	IFR	Fatigue, corrosion, aging, friction, cyclical loading.	Derating, PM, parts replacement, Technology

Conditional Reliability

Conditional probability is useful in describing the reliability of a component or system following a burn-in period T_0 or after a warranty period T_0 . We define conditional reliability of a system given that it has operated for time T_0 .

$$R(t|T_0) = Pr\{T > (T_0 + t) | T > T_0\} = \frac{Pr\{T > (T_0 + t)\}}{Pr\{T > T_0\}}$$

$$= \frac{R(T_0 + t)}{R(T_0)} = \frac{\exp\left[-\int_0^{T_0+t} \lambda(t)dt\right]}{\exp\left[-\int_0^{T_0} \lambda(t)dt\right]} = \exp\left[-\int_{T_0}^{T_0+t} \lambda(t)dt\right]$$

Example: The failure rate function $\lambda(t)$ of a system component is DFR and is given by $\frac{0.5}{1000} \left(\frac{t}{1000} \right)^{-0.5}$ when t is in years. Find the designed life of the component for a reliability value of 0.9. What improvement in design life will be if the manufacturer provides $\frac{1}{2}$ year burn-in warranty?

Constant failure rate model

A failure distribution that has a constant failure rate is an exponential probability distribution.

Exponential Reliability Function

The CFR model is developed by assuming that $\lambda(t) = \lambda$, $t \geq 0$, $\lambda > 0$

$$R(t) = \exp \left[- \int_0^t \lambda dt \right] = e^{-\lambda t} \quad t \geq 0$$

$$F(t) = 1 - R(t) = 1 - e^{-\lambda t}$$

$$f(t) = - \frac{d(R(t))}{dt} = \lambda e^{-\lambda t}$$

$$MTTF = \int_0^{\infty} R(t)dt = \int_0^{\infty} e^{-\lambda t} dt = \left. \frac{e^{-\lambda t}}{-\lambda} \right|_0^{\infty} = \frac{1}{\lambda}$$

$$\sigma^2 = \int_0^{\infty} \left(t - \frac{1}{\lambda} \right)^2 \lambda e^{-\lambda t} dt = \frac{1}{\lambda^2}$$

And the standard deviation is $\frac{1}{\lambda} = MTTF$. This is an interesting result since it implies that the variability of failure time increases as the reliability (MTTF) increases.

$$R(MTTF) = e^{-\left(\frac{MTTF}{MTTF}\right)} = e^{-1} = 0.368$$

Designed life(t_R) for a given reliability value R is

$$R(t_R) = e^{-\lambda t_R} = R \Rightarrow t_R = -\frac{1}{\lambda} \ln R$$

When $R = 0.5$, the median of the distribution $t_{med} = -\frac{1}{\lambda} \ln 0.5 = 0.69315 MTTF$

Memorylessness

A well-known characteristic of the CFR model, one not shared by other failure distribution, is its lack of memory. That is, the TTF of a component is not dependent on how long the component has been operating. There is no aging or wear-out effect. The probability that the component will operate for the next 1000 hours is the same regardless of whether the component is brand new, has been operating for several hundred hours, or has been operating for several thousand hours. This property is consistent with the completely random and independent nature of the failure process. For example, when external random environmental stresses are the primary cause of failure, the failure or operating history of the component will not be relevant.

This property can be demonstrated mathematically using conditional reliability.

$$R(t|T_0) = \frac{R(t + T_0)}{R(T_0)} = \frac{e^{-\lambda(t+T_0)}}{e^{-\lambda T_0}} = \frac{e^{-\lambda t} \cdot e^{-\lambda T_0}}{e^{-\lambda T_0}} = e^{-\lambda t} = R(t)$$

In other words, a burn-in period T_0 has no subsequent effect on reliability and will not improve the component's reliability. TTF depends only on the length of the observed operating time (t) and not on its current age (T_0).

Failure modes

$R_i(t)$ is the reliability function for the i -th failure mode, then assuming independence among the failure modes, the system reliability $R(t)$ is

$$R(t) = \prod_{i=1}^n R_i(t)$$

$R_i(t)$ = i -th failure mode does not occur before time t , and

$R(t)$ = none of the n failure modes occurs before time t .

System hazard rate or failure rate $\lambda(t)$,

Let $\lambda_i(t)$ = failure rate function for the i -th failure mode.

$$R_i(t) = \exp \left[- \int_0^t \lambda_i(t) dt \right]$$

$$R(t) = \prod_{i=1}^n \exp \left[- \int_0^t \lambda_i(t) dt \right] = \exp \left[- \int_0^t \sum_{i=1}^n \lambda_i(t) dt \right] = \exp \left[- \int_0^t \lambda(t) dt \right]$$

When $\lambda(t) = \sum_{i=1}^n \lambda_i(t)$

Example: A system is made of three components. The failure rate (number of failures per operating hours) function of these components are

$\lambda_1(t) = \frac{1}{20} \left(\frac{t}{10} \right)^{-\frac{1}{2}}$, $\lambda_2(t) = 0.01$ and $\lambda_3(t) = \frac{1}{50} \left(\frac{t}{100} \right)$. Give the system reliability function.

Failure modes with CFR Model

If a system consists of n independent, serially related components each having a constant failure rate λ_i , then

$$\lambda(t) = \lambda = \sum_{i=1}^n \lambda_i \quad \text{and}$$

$$R(t) = \exp \left[- \int_0^t \lambda dt \right] = \exp[-\lambda t]$$

$$\text{Where } MTTF = \frac{1}{\lambda} = \frac{1}{\sum_{i=1}^n \lambda_i} = \frac{1}{\sum_{i=1}^n \frac{1}{MTTF_i}} \quad MTTF_i = \frac{1}{\lambda_i}$$

In other words, the system itself will have an exponential time CFR model. If the components are also all identical, i.e., $\lambda_i = \lambda_1$, for $i = 2, 3, \dots, n$, then $\lambda = n\lambda_1$ and $MTTF = 1/(n\lambda_1)$

Example: An aircraft engine consists of three modules having constant failure rates of $\lambda_1 = 0.002$, $\lambda_2 = 0.015$ and $\lambda_3 = 0.0025$ failures per operating hour. Give the reliability function for the engine and its MTTF.

Failures on Demand

Components operate on cyclical basis may fail on demand. For example, an air conditioning and heating system whose cycle time includes an operating time and an idle time may fail when switching from an idle to an operating mode.

λ_I = the average failure rate while idle

λ_O = the average failure rate while operating

p = probability of failure on demand

t_I = average length of idle time period per cycle

t_O = average length of the operating time period per cycle

Failure rate(failures per clock unit of time), $\lambda_{eff} = \frac{t_I}{t_I+t_O} \times \lambda_I + \frac{t_O}{t_I+t_O} \times \lambda_O + \frac{p}{t_I+t_O}$

Example: An air conditioning compressor operates once for an operating time of 20 minutes each hour. While operating, it has experienced a failure rate of 0.01 failures per operating hour and while idle has experienced a dormant failure rate of 0.0002 failure per idle hour. The probability that the compressor fails on demand is 0.03. What is the probability that the compressor will not fail over a 24-hour period?

Two parameter exponential distribution

If a failure will never occur prior to some specified time t_0 , then t_0 is a **minimum or threshold, time**. It is also known as the guaranteed lifetime. The parameter t_0 is a location parameter that shifts the distribution an amount equal to t_0 to the right on the time axis. This is equivalent to rewriting the density function by replacing t with $t-t_0$ with domain of random variable now $t \geq t_0$. For the exponential distribution, the PDF becomes,

$$f(t) = \frac{-d(R(t))}{dt} = \lambda e^{-\lambda(t-t_0)} \quad 0 < t_0 \leq t < \infty$$

$$\text{Reliability, } R(t) = e^{-\lambda(t-t_0)} \quad t \geq t_0$$

Failure rate, $\lambda(t) = \frac{f(t)}{R(t)} = \lambda$

Mean of the distribution

$$MTTF = \int_{t_0}^{\infty} \lambda t e^{-\lambda(t-t_0)} dt = t_0 + \frac{1}{\lambda}$$

Median of the distribution t_{med} ,

$$R(t_{med}) = e^{-\lambda(t_{med}-t_0)} = 0.5 \Rightarrow t_{med} = t_0 + \frac{\ln 0.5}{-\lambda}$$

The designed life for a specified reliability value R

$$t_R = t_0 + \frac{\ln R}{-\lambda}$$

Variance and S. D. of two parameter exponential distribution are not affected by location parameter.

Therefore variance, $\sigma^2 = \frac{1}{\lambda^2}$; and S.D, $\sigma = \frac{1}{\lambda}$ and mode = t_0

Example: A component experiences 0.001 failures per hour. It has a guaranteed life of 200 hours. Find the reliability at 300 hours of operation. Calculate MTTF and σ . What is the median life of the component? And calculate the designed life if a 0.95 reliability is desired.

Poisson Process

If a component having a constant failure rate λ is immediately repaired and replaced upon failing, the number of failures observed over a time period t has a Poisson distribution. The probability of observing n failures in time t is given by the Poisson probability mass function $p_n(t)$.

$$p_n(t) = \frac{e^{-\lambda t} (\lambda t)^n}{n!} \quad \text{when } n = 0, 1, 2, \dots \dots$$

Expected number of failures over time t is λt and the variance is λt .

Probability of zero failures occurring in time t is

$$Pr\{T \geq t\} = p_0(t) = \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t} = R(t)$$

The Poisson process is often used in inventory analysis to determine the number of spare components when the time between failures is exponential. For example, if S spare components are available to support a continuous operation over time period t , then

$$R_S(t) = \sum_{n=0}^S p_n(t)$$

is the cumulative probability of S or fewer failures occurring during time t . Therefore, $R_S(t)$, is the component reliability if there are S spares available for immediate replacement when a failure occurs.

Example: A specially designed welding machine has a nonrepairable motor with a constant failure rate of 0.05 failures per year. The company has purchased two spare motors. If the designed life of the welding machine is 10 years, what is the probability that two spares will be adequate?

Redundancy and the CFR Model

Consider the case of two independent and redundant components each having the same constant failure rate λ . In this case a system failure will occur when both the components have failed. Since $(1 - e^{-\lambda t})^2$, is the probability that both components will fail by time t , the system reliability $R(t)$ is given by

$$R(t) = 1 - (1 - e^{-\lambda t})^2 = 1 - (1 - 2e^{-\lambda t} + e^{-2\lambda t}) = 2e^{-\lambda t} - e^{-2\lambda t}$$

And the pdf is given by,

$$f(t) = \frac{-d(R(t))}{dt} = -[-2\lambda e^{-\lambda t} + 2\lambda e^{-2\lambda t}] = [2\lambda e^{-\lambda t} - 2\lambda e^{-2\lambda t}]$$

The hazard rate in this case,

$$\lambda(t) = \frac{f(t)}{R(t)} = \frac{2\lambda e^{-\lambda t} - 2\lambda e^{-2\lambda t}}{2e^{-\lambda t} - e^{-2\lambda t}} = \frac{\lambda(1 - e^{-\lambda t})}{(1 - 0.5e^{-\lambda t})}$$

The system MTTF can be determined from,

$$\begin{aligned} MTTF &= \int_0^{\infty} R(t).dt = \int_0^{\infty} (2e^{-\lambda t} - e^{-2\lambda t}) dt = \left(\frac{2e^{-\lambda t}}{-\lambda} - \frac{e^{-2\lambda t}}{-2\lambda} \right) \Bigg|_0^{\infty} \\ &= \left(-\frac{2}{\lambda e^{\lambda t}} + \frac{1}{2\lambda e^{2\lambda t}} \right) \Bigg|_0^{\infty} = \frac{2}{\lambda} - \frac{1}{2\lambda} = \frac{3}{2\lambda} = \frac{1.5}{\lambda} \end{aligned}$$